3. Exercise

Linear Models and Regression

Exercise Sheet 3

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Solve the following problems before the exercise class on November 20.

Problem 1
It is assumed that the mortality caused by malignant melanomas depends on the sunshine intensity, which corresponds to the latitude. The following table provides for several states in the US the number of deaths per 10 millions of the white population between 1950 and 1969 as well as the latitude.

State	Delaware	Iowa	Michigan	New Hampshire	Oklahoma	Texas	Wyoming
Mortality	200	128	117	129	182	229	134
Latitude	39	42	44	44	35	31	43

In the following, linear regression is used to explain the mortality by the latitude.

a) Estimate the intercept and the coefficient of the linear model as well as the variance of the random errors.

b) Predict the mortality for Ohio whose latitude is 40.

From now on we assume a Gaussian linear model.

c) Use a statistical test with level 0.05 to show that the slope is negative.

d) Derive a confidence interval for the mortality in Ohio with error level 0.1.

For testing hypotheses and computing confidence intervals one can use the functions `glht`, `summary.glht` and `confint.glht` from the `R` package `multcomp`. The residuals of a linear model `LM` can be accessed by `LM$residuals`.

e) Solve the problems a) to d) by using `R`.

Problem 2
In order to check the assumption of the Gaussian linear model that the error terms are i.i.d. normally distributed, one can use the following graphical tools:

- In a normal qq-plot, the empirical quantiles of the residuals are plotted against the quantiles of the standard normal distribution. If the points form approximatively a straight line, this indicates that the residuals are normally distributed. The command for a normal qq-plot in `R` is `qqnorm`.

#Problem 1

Recall

$$Y = \gamma_0 + \gamma_1 x^{(1)} + \dots + \gamma_k x^{(k)} + \varepsilon$$

$\Updownarrow$

1st observation

$$Y_1 = \gamma_0 + \gamma_1 x_1^{(1)} + \dots + \gamma_k x_1^{(k)} + \varepsilon_1$$

$$\vdots$$

$$Y_n = \gamma_0 + \gamma_1 x_n^{(1)} + \dots + \gamma_k x_n^{(k)} + \varepsilon_n$$

$\Updownarrow$

$$Y = A\gamma + \varepsilon$$

design matrix

$$A = \begin{pmatrix} 1 & x_1^{(1)} & \dots & x_1^{(k)} \\ \vdots & \vdots & & \\ 1 & x_n^{(k)} & \dots & x_n^{(k)} \end{pmatrix}$$

$$\gamma = \begin{pmatrix} \gamma_0 \\ \vdots \\ \gamma_k \end{pmatrix} \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

$\Rightarrow$

a) Dependent variable: mortality
independent variable: latitude

$\Downarrow$

$$\text{mortality} = \gamma_0 + \gamma_1 \cdot \text{Latitude} + \varepsilon$$

We have

$$A = \begin{pmatrix} 1 & 39 \\ 1 & 42 \\ 1 & 44 \\ 1 & 44 \\ 1 & 35 \\ 1 & 31 \\ 1 & 43 \end{pmatrix} \quad Y = \begin{pmatrix} 200 \\ 128 \\ 117 \\ 129 \\ 182 \\ 229 \\ 134 \end{pmatrix}$$

LSE (Least Square Estimator)

$$\hat{\gamma} = (A^T A)^{-1} A^T Y$$

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 7 & 278 \\ 278 & 11192 \end{pmatrix}$$

$$A^T Y = \begin{pmatrix} 1119 \\ 43231 \end{pmatrix}$$

$$(A^T A)^{-1} = \frac{1}{7 \cdot 11192 - 278 \cdot 278} \begin{pmatrix} 11192 & -278 \\ -278 & 7 \end{pmatrix}$$

$$= \frac{1}{1060} \begin{pmatrix} 11192 & -278 \\ -278 & 7 \end{pmatrix}$$

So,

$$\hat{\gamma} = \frac{1}{1060} \begin{pmatrix} 11192 & -278 \\ -278 & 7 \end{pmatrix} \begin{pmatrix} 1119 \\ 43231 \end{pmatrix}$$

$$\boxed{\hat{\gamma} \approx \begin{pmatrix} 477.01 \\ -7.99 \end{pmatrix}} \begin{matrix} \rightarrow \hat{\gamma}_0 \text{ (intercept)} \\ \rightarrow \hat{\gamma}_1 \text{ (coefficient)} \end{matrix}$$

We have

$$\hat{\varepsilon} = Y - A\hat{\gamma} = \begin{pmatrix} 34.6 \\ -13.43 \\ -8.45 \\ 3.55 \\ -15.36 \\ -0.32 \\ 0.56 \end{pmatrix}$$

Recall

$$\hat{\sigma}^2 = \frac{1}{n-k-1} \|\hat{\varepsilon}\|^2$$

where $\hat{\varepsilon} = Y - \hat{Y} = Y - A\hat{\gamma}$

$$\therefore \hat{\sigma}^2 = \frac{1}{5} 1697.88 = 339.58 \qquad [k=1]$$

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b) The prediction for Ohio is

$$477.01 - 7.99 \times 40 = \underline{157.41}$$

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c)

Recall

$(\gamma_0, \gamma_1, \gamma_2)$

$H_0: \gamma_2 > 0 \quad ; \quad H_1: \gamma_2 \leq 0$

$$\gamma_2 = \underbrace{(0, 0, 1)}_{a} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \gamma_2 \end{pmatrix}$$

$$= \langle a, \gamma \rangle$$

$H_0: \langle a, \gamma \rangle > c_0 \quad \text{vs} \quad \langle a, \gamma \rangle \leq c_0$ $\quad c_0 = 0$

i) $H_0: \langle a, \gamma \rangle = c_0 \quad \text{vs} \quad H_1: \langle a, \gamma \rangle \neq c_0$

$$\varphi_{\neq}(\gamma) = \begin{cases} \text{cannot reject} & c_0 \in I^{\neq}_{a,\hat{\gamma}} \\ \text{reject} & c_0 \notin I^{\neq}_{a,\hat{\gamma}} \end{cases}$$

$$I^{\neq}_{a,\hat{\gamma}} = \begin{pmatrix} \langle a, \hat{\gamma} \rangle - t_{n-k-1, 1-\frac{\alpha}{2}} \hat{\sigma}\sqrt{a^T(A^TA)^{-1}a} \\ \langle a, \hat{\gamma} \rangle + t_{n-k-1, 1-\frac{\alpha}{2}} \hat{\sigma}\sqrt{a^T(A^TA)^{-1}a} \end{pmatrix}$$

b) $H_0 : \langle a, r \rangle \leq C_0$ vs $H_1 : \langle a, r \rangle > C_0$

$\left(\underset{1-\alpha}{1^{st}} , \infty \right)$

c) $H_0 : \langle a, r \rangle \geq C_0$ vs $H_1 : \langle a, r \rangle < C_0$

$\left(-\infty , \underset{1-\alpha}{2^{nd}} \right)$

$H_0 : r_1 \geq 0$ vs $H_1 : r_1 < 0$

we have $a = (0, 1)$, $C_0 = 0$

$$\hat{r}_1 + t_{5, 0.95} \cdot \hat{\sigma} \sqrt{\left((A^T A)^{-1} \right)_{2,2}} \rightarrow 2^{nd} \text{ row}, 2^{nd} \text{ column} = \boxed{-5.01}$$

$$a^T (A^T A) a = (0 \ 1) \begin{pmatrix} 7 & 278 \\ 278 & 11192 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 11192$$

$0 \notin (-\infty, -5.01)$

We can reject H_0 at level 0.05

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d) The confidence interval : $a = (1, 40)$

$$\left(\hat{r}_0 + 40 \hat{r}_1 - t_{5, 0.95} \hat{\sigma} \sqrt{(1 \ 40) (A^T A)^{-1} \begin{pmatrix} 1 \\ 40 \end{pmatrix}} , \hat{r}_0 + 40 \hat{r}_1 + t_{5, 0.95} \hat{\sigma} \sqrt{(1 \ 40) (A^T A)^{-1} \begin{pmatrix} 1 \\ 40 \end{pmatrix}} \right)$$

$= (143.48 , 171.34)$

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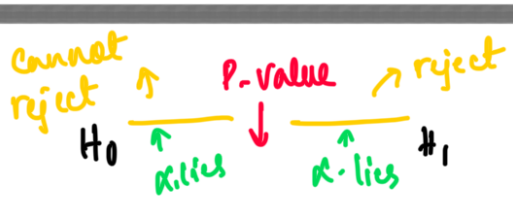
e)



Problem 3

$n_1, n_2 \in \mathbb{N}$

$z = (z_i^{(1)} \quad z_i^{(n_1)})^T$

$$Z_1 = (z_1^{(1)} \dots z_1^{(n_1)})^T$$

$$Z_2 = (z_2^{(1)} \dots z_2^{(n_2)})^T$$

$$\mathcal{N}(\mu_1, \sigma^2) \qquad \mathcal{N}(\mu_2, \sigma^2)$$

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$$

$$0 \notin \left(\bar{z}_2 - \bar{z}_1 - \frac{\sqrt{n_1+n_2}}{\sqrt{n_1 n_2}} t_{n_1+n_2-2,\, 1-\frac{\alpha}{2}} , \right.$$

$$\left. \bar{z}_2 - \bar{z}_1 + \frac{\sqrt{n_1+n_2}}{\sqrt{n_1 n_2}} t_{n_1+n_2-2,\, 1-\frac{\alpha}{2}} \right)$$

a) Let $Y = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$

$$A = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{pmatrix} \begin{matrix} \left.\vphantom{\begin{matrix}1\\1\\1\end{matrix}}\right\} n_1\text{-times} \\ \left.\vphantom{\begin{matrix}1\\1\\1\end{matrix}}\right\} n_2\text{-times} \end{matrix} \qquad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_{n_1+n_2} \end{pmatrix}$$

Then

$$Y = A\gamma + \varepsilon \quad \text{with}$$

$$\gamma = \begin{pmatrix} \gamma_0 \\ \gamma_1 \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 - \mu_1 \end{pmatrix}$$

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b) We have

$$A^T A = \begin{pmatrix} 1 \dots\dots 1 \\ 0 \dots 0 1 \dots 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} n_1+n_2 & n_2 \\ n_2 & n_2 \end{pmatrix}$$

$$A^T Y = \begin{pmatrix} 1 \dots\dots 1 \\ 0 \dots 0 1 \dots 1 \end{pmatrix} \begin{pmatrix} z_1^{(1)} \\ \vdots \\ z_1^{(n_1)} \\ z_2^{(1)} \end{pmatrix}$$

$$\begin{pmatrix} \vdots \\ z_2^{(n_2)} \end{pmatrix}$$

$$= \begin{pmatrix} n_1 \bar{z}_1 + n_2 \bar{z}_2 \\ n_2 \bar{z}_2 \end{pmatrix}$$

$$(A^T A)^{-1} = \frac{1}{n_2(n_1+n_2) - n_2^2} \begin{pmatrix} n_2 & -n_2 \\ -n_2 & n_1+n_2 \end{pmatrix}$$

$$= \frac{1}{n_1 n_2} \begin{pmatrix} n_2 & -n_2 \\ -n_2 & n_1+n_2 \end{pmatrix}$$

$$\hat{\gamma} = (A^T A)^{-1} A^T Y = \frac{1}{n_1 n_2} \begin{pmatrix} n_2 & -n_2 \\ -n_2 & n_1+n_2 \end{pmatrix} \begin{pmatrix} n_1 \bar{z}_1 + n_2 \bar{z}_2 \\ n_2 \bar{z}_2 \end{pmatrix}$$

$$= \frac{1}{n_1 n_2} \begin{pmatrix} n_1 n_2 \bar{z}_1 \\ n_1 n_2 (\bar{z}_2 - z_1) \end{pmatrix}$$

$$= \begin{pmatrix} \bar{z}_1 \\ \bar{z}_2 - \bar{z}_1 \end{pmatrix}$$

$$\hat{\varepsilon} = Y - A\hat{\gamma} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} - \begin{pmatrix} \bar{z}_1 \\ \vdots \\ \bar{z}_1 \\ \bar{z}_2 \\ \vdots \\ \bar{z}_2 \end{pmatrix}$$

$$\|\hat{\varepsilon}\|^2 = \sum_{i=1}^{n_1} (z_1^{(i)} - \bar{z})^2 + \sum_{i=1}^{n_2} (z_2^{(i)} - \bar{z}_2)^2$$

$$\sigma^2 = \frac{1}{n_1+n_2-2} \|\hat{\varepsilon}\|^2$$

$H_0: \gamma_1 = 0$ vs $\gamma_1 \neq 0$

$$\left(\hat{\gamma}_1 - t_{1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{\left((A^T A)^{-1}\right)_{2,2}} \;,\; \hat{\gamma}_1 + t_{1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{\left((A^T A)^{-1}\right)_{2,2}}\right)$$

$$= \left(\bar{z}_2 - \bar{z}_1 - t_{1-\frac{\alpha}{2}}\right.$$